THE HOMOTOPY CATEGORY OF AN ADDITIVE CATEGORY IS TRIANGULATED

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3545-3603, following the statement at 3536-3543.

Editorial note. TR3 and TR4 are instantiation chains of same-chapter lemmas with complete proofs (functorial-cone; easier-axiom-four, sequence-maps-split already carded, two-split-injections queued); the reduction data is written out, recorded.

Allowed standard prerequisites (editorial).

- Cone triangles represent distinguished triangles. lemma-the-same-up-to-isomorphisms (carded in this batch, L26170), cited at proof lines 3551, 3571, 3573, 3588. Supplies that distinguished triangles of K(A) are exactly the triangles isomorphic to cone triangles, used to reduce TR1/TR2/TR3 to cone data.
- Rotation of cone triangles. lemma-rotate-triangle and lemma-rotate-cone, cited at proof lines 3563 and 3579; short source lemmas with complete proofs in the same chapter. Supplies the rotated-triangle identifications (with the sign bookkeeping) in the two directions of TR2.
- Functorial cone. lemma-functorial-cone, cited at proof line 3591; complete short proof at source lines 2736-2753. Supplies the morphism of cone triangles in TR3.
- Reduction of TR4 for pre-triangulated categories. lemma-easier-axiom-four, cited at proof line 3597; complete proof at source lines 1064-1074. Supplies that TR4 follows from the composable-pair octahedron statement.
- Lifting to split injections in K(A). lemma-sequence-maps-split (carded L26103), cited at proof line 3599. Supplies that the composable pair may be represented by split injective maps of complexes.
- Octahedron for split injections. lemma-two-split-injections, cited at proof line 3602; complete proof at source lines 3485-3534, queued for separate carding. Supplies the octahedron in the split-injection case, finishing TR4.

Source reference. Original lemma 014C: sources/stacks-derived/pinned-original-source.tex, source lines 1046–1046. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014F: sources/stacks-derived/pinned-original-source.tex, source lines 2718–2718. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014O: sources/stacks-derived/pinned-original-source.tex, source lines 3424–3424. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014N: sources/stacks-derived/pinned-original-source.tex, source lines 3393–3393. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014M: sources/stacks-derived/pinned-original-source.tex, source lines 3357–3357. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014L: sources/stacks-derived/pinned-original-source.tex, source lines 3213–3213. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014R: sources/stacks-derived/pinned-original-source.tex, source lines 3474–3474. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $\mathcal{A}$ be an additive category. The category $K(\mathcal{A})$ of complexes up to homotopy with its natural translation functors and distinguished triangles as defined above is a triangulated category.*

Proof. Proof of TR1. By definition every triangle isomorphic to a distinguished one is distinguished. Also, any triangle $(A^\bullet, A^\bullet, 0, 1, 0, 0)$ is distinguished since $0 \to A^\bullet \to A^\bullet \to 0 \to 0$ is a termwise split sequence of complexes. Finally, given any morphism of complexes $f : K^\bullet \to L^\bullet$ the triangle $(K, L, C(f), f, i, -p)$ is distinguished by Lemma 014L.

Proof of TR2. Let (X, Y, Z, f, g, h) be a triangle. Assume $(Y, Z, X[1], g, h, -f[1])$ is distinguished. Then there exists a termwise split sequence of complexes $A^\bullet \to B^\bullet \to C^\bullet$ such that the associated triangle $(A^\bullet, B^\bullet, C^\bullet, \alpha, \beta, \delta)$ is isomorphic to $(Y, Z, X[1], g, h, -f[1])$. Rotating back we see that (X, Y, Z, f, g, h) is isomorphic to $(C^\bullet[-1], A^\bullet, B^\bullet, -\delta[-1], \alpha, \beta)$. It follows from Lemma 014N that the triangle $(C^\bullet[-1], A^\bullet, B^\bullet, \delta[-1], \alpha, \beta)$ is isomorphic to $(C^\bullet[-1], A^\bullet, C(\delta[-1])^\bullet, \delta[-1], i, p)$. Precomposing the previous isomorphism of triangles with -1 on Y it follows that (X, Y, Z, f, g, h) is isomorphic to $(C^\bullet[-1], A^\bullet, C(\delta[-1])^\bullet, \delta[-1], i, -p)$. Hence it is distinguished by Lemma 014L. On the other hand, suppose that (X, Y, Z, f, g, h) is distinguished. By Lemma 014L this means that it is isomorphic to a triangle of

the form $(K^\bullet, L^\bullet, C(f), f, i, -p)$ for some morphism of complexes f. Then the rotated triangle $(Y, Z, X[1], g, h, -f[1])$ is isomorphic to $(L^\bullet, C(f), K^\bullet[1], i, -p, -f[1])$ which is isomorphic to the triangle $(L^\bullet, C(f), K^\bullet[1], i, p, f[1])$. By Lemma 014O this triangle is distinguished. Hence $(Y, Z, X[1], g, h, -f[1])$ is distinguished as desired.

Proof of TR3. Let (X, Y, Z, f, g, h) and (X', Y', Z', f', g', h') be distinguished triangles of $K(\mathcal{A})$ and let $a : X \to X'$ and $b : Y \to Y'$ be morphisms such that $f' \circ a = b \circ f$. By Lemma 014L we may assume that $(X, Y, Z, f, g, h) = (X, Y, C(f), f, i, -p)$ and $(X', Y', Z', f', g', h') = (X', Y', C(f'), f', i', -p')$. At this point we simply apply Lemma 014F to the commutative diagram given by f, f', a, b.

Proof of TR4. At this point we know that $K(\mathcal{A})$ is a pre-triangulated category. Hence we can use Lemma 014C. Let $A^\bullet \to B^\bullet$ and $B^\bullet \to C^\bullet$ be composable morphisms of $K(\mathcal{A})$. By Lemma 014M we may assume that $A^\bullet \to B^\bullet$ and $B^\bullet \to C^\bullet$ are split injective morphisms. In this case the result follows from Lemma 014R. □